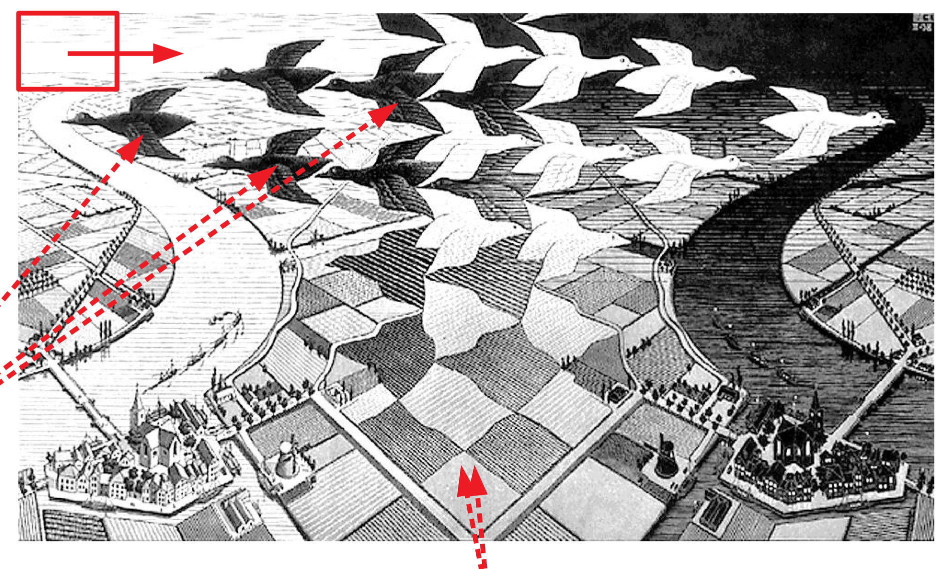
An extension of convolutional neural networks to irregular domains through graph inference and translations identification

Bastien Pasdeloup
EPFL
bastien.pasdeloup@epfl.ch

Jean-Charles Vialatte
IMT Atlantique
jc.vialatte@imt-atlantique.fr

Vincent Gripon
IMT Atlantique
vincent.gripon@imt-atlantique.fr

CNNs capture these properties by translating a localized kernel

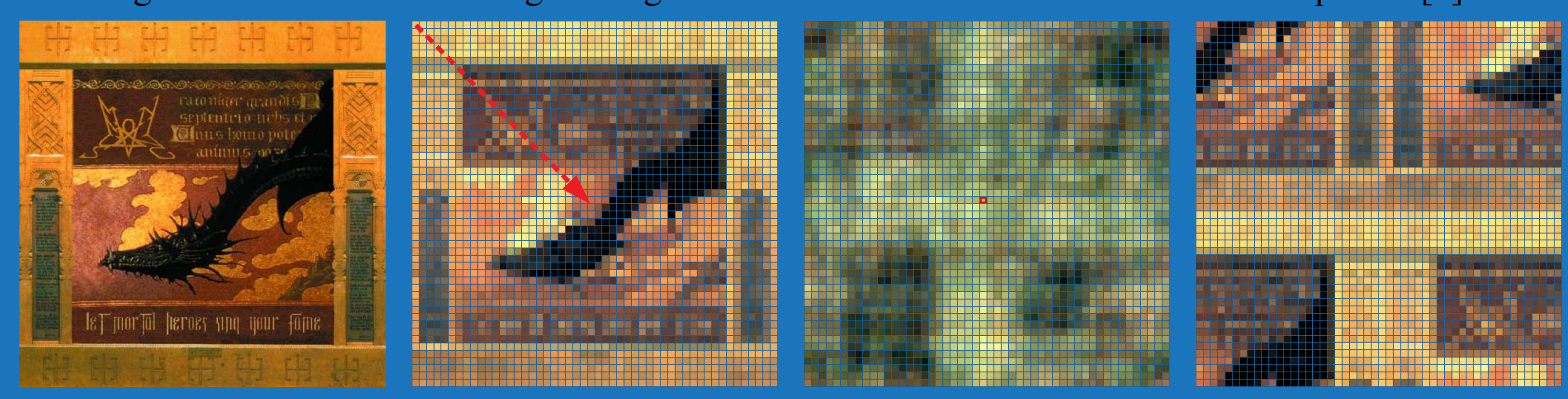


Locations of the objects should not matter (*stationarity* of the domain) [2,3]

Close pixels share similar values (*smoothness* of objects) [1]

Main assumption :

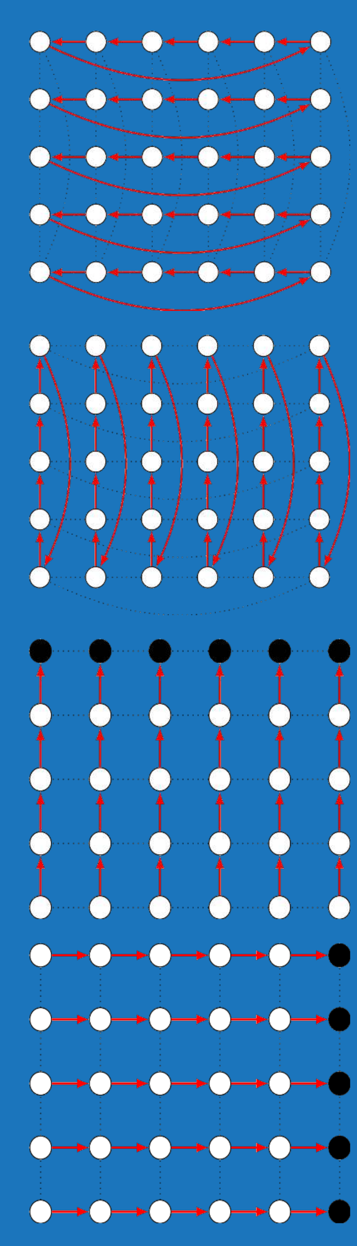
In order to be identifiable by a graph CNN, objects of interest should be localized on a graph, and should remain similar at any location.

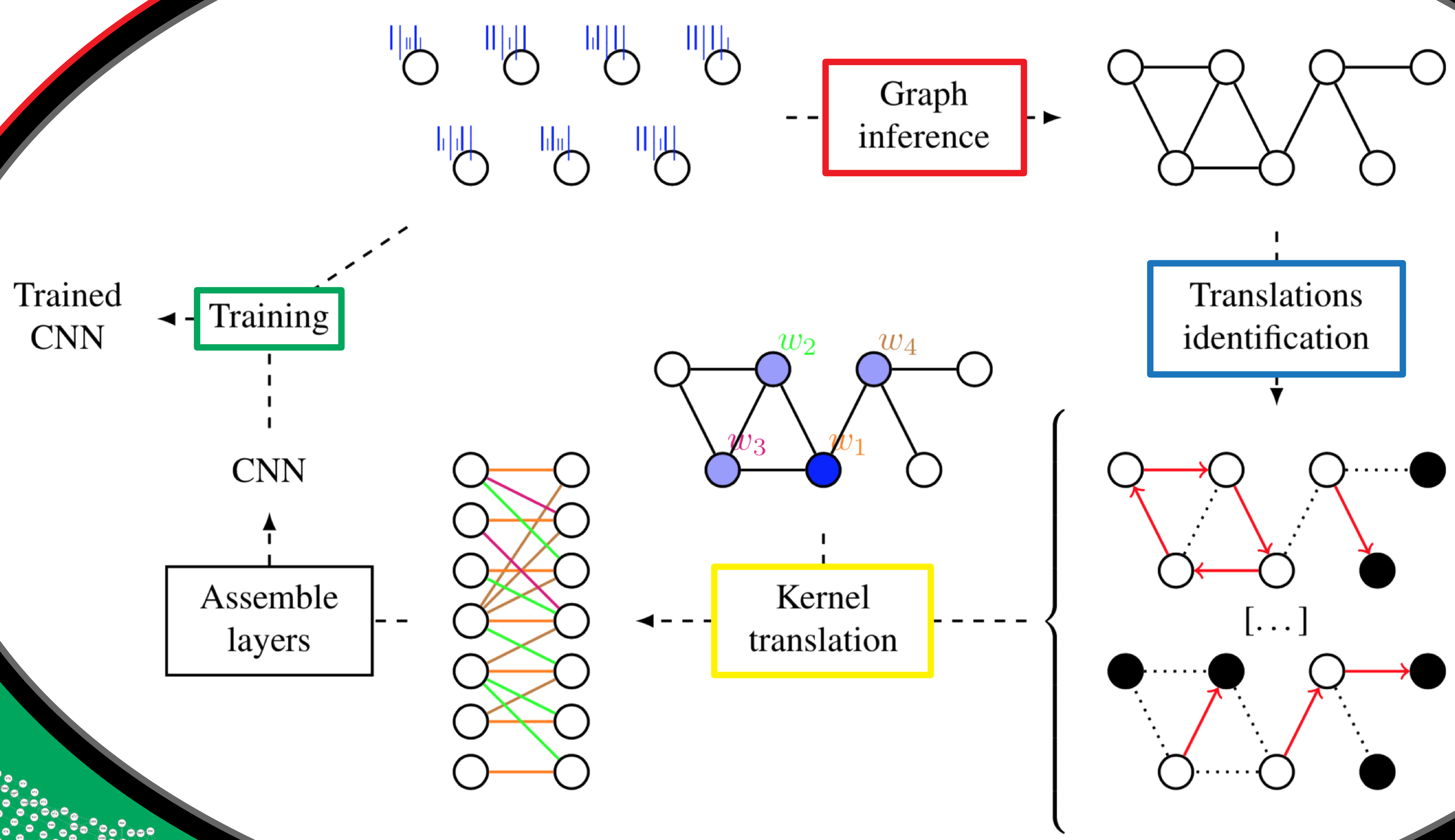


$$\forall v_1, v_2 \in \mathcal{V} : (\psi(v_1) = \psi(v_2) \neq \bot) \Rightarrow (v_1 = v_2)$$

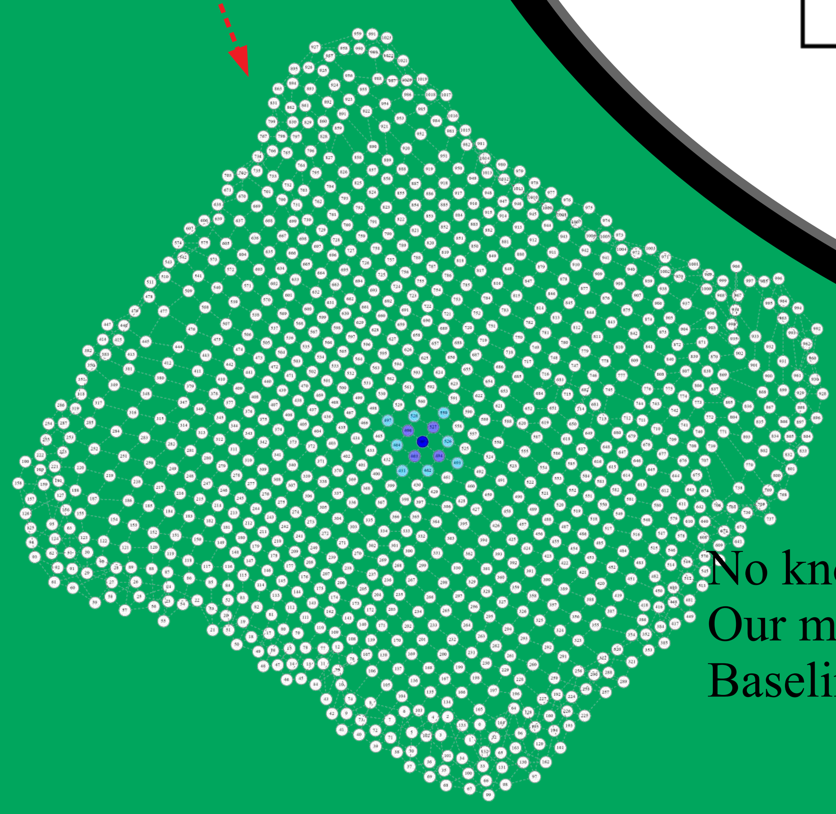
$$\forall v \in \mathcal{V} : (\{v, \psi(v)\} \in \mathcal{E}) \vee (\psi(v) = \bot)$$

$$\forall v_1, v_2 \in \mathcal{V} : (\{v_1, v_2\} \in \mathcal{E} \Leftrightarrow \{\psi(v_1), \psi(v_2)\} \in \mathcal{E}) \vee (\psi(v_1) = \bot) \vee (\psi(v_2) = \bot)$$



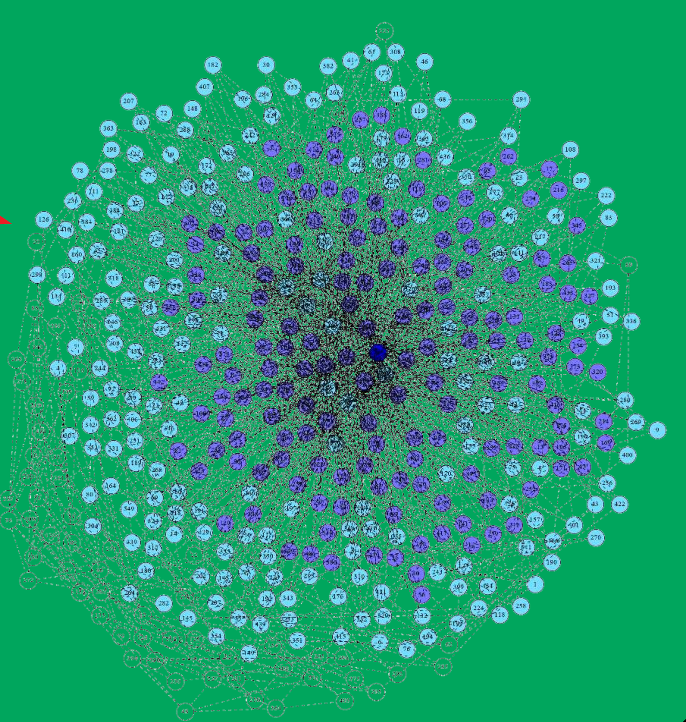


Scrambled CIFAR-10

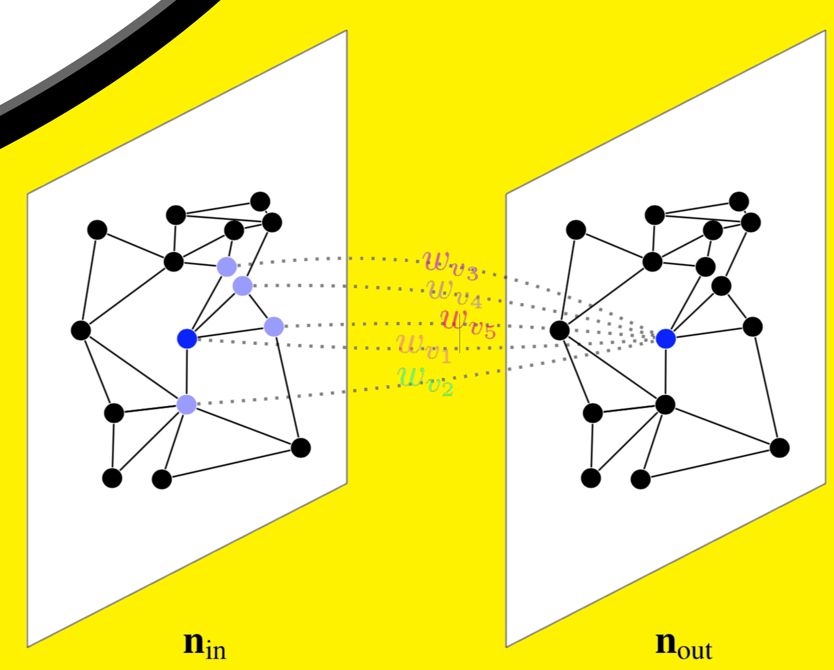


No knowledge on topology [5]: 72.70 %
Our method (simple CNN arch.): 85.52 %
Baseline (Same arch., 3x3 filters): 85.98 %

Haxby (fMRI)



	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5	Subject 6	Average
Our method with $\mathcal{G}_{geo}$	56%	52%	47%	53%	42%	49%	49.9%
$\text{Def}_5(\mathcal{G}_{geo})$ [6]	60%	**56%**	**51%**	54%	**45%**	51%	52.8%
$\text{Def}_{10}(\mathcal{G}_{geo})$ [6]	61%	55%	50%	55%	42%	51%	52.3%
$\text{Def}_{20}(\mathcal{G}_{geo})$ [6]	**62%**	54%	**51%**	**56%**	43%	**53%**	**53.2%**
$\text{Def}_5(\mathcal{G}^*_K)$ [6]	61%	53%	50%	53%	43%	49%	51.5%
$\text{Def}_{10}(\mathcal{G}^*_K)$ [6]	59%	54%	50%	54%	**45%**	51%	52.2%
$\text{Def}_{20}(\mathcal{G}^*_K)$ [6]	61%	51%	49%	54%	42%	49%	51%
MLP	57%	51%	**51%**	51%	42%	49%	50.2%
SVM	50%	42%	35%	32%	31%	40%	38.2%
LR ($C = 1$, l_1)	51%	48%	44%	34%	42%	42%	43.6%
LR ($C = 50$, l_1)	47%	47%	38%	28%	34%	40%	38.9%
LR ($C = 1$, l_2)	49%	47%	41%	34%	31%	38%	39.7%
Ridge	35%	34%	28%	22%	23%	31%	28.8%



$$\begin{aligned} S(\widetilde{\psi}) \;=\; & \alpha|\{v \in \mathcal{V}_{(v_1,P)} : \widetilde{\psi}(v) = \bot\}| \\ +\; & \beta|\{v \in \mathcal{V}_{(v_1,P)} : (v, \widetilde{\psi}(v)) \notin \mathcal{E} \wedge \widetilde{\psi}(v) \neq \bot\}| \\ +\; & \gamma|\{(v_1, v_2) \in \mathcal{V}^2_{(v_1,P)} : \\ & \Big(\Big(((v_1,v_2) \in \mathcal{E}) \wedge ((\widetilde{\psi}(v_1), \widetilde{\psi}(v_2)) \notin \mathcal{E}\Big) \vee \\ & \Big(((v_1,v_2) \notin \mathcal{E}) \wedge ((\widetilde{\psi}(v_1), \widetilde{\psi}(v_2)) \in \mathcal{E}\Big)\Big) \\ & \wedge (\widetilde{\psi}(v_1) \neq \bot) \wedge (\widetilde{\psi}(v_2) \neq \bot)\}| \end{aligned}$$

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[2] B.Pasdeloup, V.Gripon, G.Mercier, D.Pastor, and M.G.Rabbat, "Characterization and inference of graph diffusion processes from observations of stationary signals," IEEE Transactions on Signal and Information Processing over Networks, 2017.

[3] B. Girault, "Stationary Graph Signals using an Isometric Graph Translation," in Eusipco, Nice, France, Aug. 2015, pp. 1531–1535. [Online]. Available: https://hal.inria.fr/hal-01155902.

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[5] X. Cheng, X. Chen, and S. Mallat, "Deep haar scattering networks," Information and Inference: A Journal of the IMA, vol. 5, no. 2, pp. 105–133, 2016.

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IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

LTS4

EPFL
ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE